

Additive Manufacturing Angled Wall Study and Vector Controlled Laser System Characterization

STATEMENT OF WORK

1. Objective/Requirements

NASA Marshall Space Flight Center (MSFC) requires the ability to conduct a number of laboratory-based experimental tasks that build upon prior work completed under NASA award 80NSSC22PA055 and 80NSSC23PC201. The University of Texas-El Paso (UTEP) has several unique capabilities that distinguish it from other institutions in additive manufacture (AM) parameter development, unique machines with beam shaping lasers, a broad range of material characterization facilities, unique test facilities, and subject matter experts familiar with NASA technical requirements. These stated capabilities are not common to all AM institutions and this procurement intends to leverage experience gained. These data and specimens are needed for application into Game Changing Technology (GCD) “*Additive Manufacturing for Propulsion*” and Space Launch System (SLS) Liquid Engines Office (LEO) projects.

2. Characteristics, Scope, and Specs

The vendor shall provide development and analysis of samples with the following requirements:

Task 1: Evaluation of Inconel 718 L-PBF Angled Wall Samples (50% cost share)

The objective of this task is to determine if there is a difference in mechanical properties across 16 different machine configurations in addition to the build angles (0, 30, 45 degrees).

- UTEP to complete full characterization and microstructure of Inconel 718 L-PBF AM with various wall angles and machine configurations.
- UTEP to analysis of tested Inconel 718 L-PBF samples.
- UTEP shall evaluate samples post-mechanical testing and complete summary of data in addition to fractography.
- UTEP to provide final report (in form of journal article).

Task 2: Complete Characterization of Inconel 718 LP-DED Angled Wall Samples (50% cost share)

The objective of this task is to determine if there is an impact to the mechanical properties or microstructure from variations in build parameters and geometry from LP-DED processing.

- UTEP to complete full characterization and microstructure of Inconel 718 LP-DED AM with various wall angles and machine configurations.
- UTEP to analysis of tested Inconel 718 L-PBF samples.
- UTEP shall evaluate samples post-mechanical testing and complete summary of data in addition to fractography.
- UTEP to provide final report (in form of journal article).

Task 3: Comparison of LP-DED JBK-75 Powder Vendors (50% cost share)

The objective of this task is to determine if there is a significant difference in microstructure or properties due to changes in powder vendor for LP-DED JBK-75.

- UTEP to complete characterization of bulk and thin wall LP-DED JBK-75 built at RPMI using powder from PAC and HMI.
- NASA MSFC shall provide all samples.
- UTEP to complete characterization and analysis of tested LP-DED JBK-75 samples to understand any effects of varying powder vendors.
- UTEP to complete microstructure evaluation of LP-DED JBK-75 samples.
- UTEP shall evaluate samples post-mechanical testing and complete summary of data in addition to fractography.
- UTEP to provide final report (in form of journal article) and summary slides – similar to what has been provided already.

Task 4: Comparison of Laser Systems – Standard and Beam Shaping (50% cost share)

The objective of this task is to provide a cursory set of data comparing the Aconity beam shaping laser system and differences between a standard fixed focus laser system.

- UTEP to build samples on beam shaping system with agreed upon alloy and samples in non-beam shaping system with identical powder pedigree.
- UTEP to complete characterization and analysis of samples to compare differences of similarities between systems, specifically microstructure.
- UTEP shall evaluate samples post-mechanical testing and complete summary of data in addition to fractography.
- UTEP to provide final report and summary slides.

3. Place of Performance

UTEP Keck Center

4. Period of Performance

The tasks shall be conducted over the course of twelve months, with the duration beginning upon initiation of contract award.

5. Deliverables

1. Report from Task 1 in the form of a journal article.
2. Report from Task 2 in the form of a journal article.
3. Report from Task 3 in the form of a journal article.
4. Report from Task 4.